Supplementary Information for

Rising biocapacity total can mask ecological-capital land drawdown: a two-land model

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Equation numbers without the prefix S refer to the main text.

How to use this supplement. Section S1 defines symbols, units, and model extensions. Section S2 derives the composition identity, reduced balance branch, delay structure, and debt bounds. Section S3 documents the solver and sensitivity checks; Section S4 expands the two-land results, drawdown thresholds, and restoration comparisons. Section S5 records the NFA data, crop proxy, observation limits, and regeneration context. Section S6 covers policy extensions, Section S7 keeps the one-stock comparator separate, Section S8 reports the matched technology-wave experiment, and Section S9 adds a finite demographic-delay scan. The final two tables gather that scan and selected reference values for easier cross-checking.

S1 Symbols and model extensions

A global hectare (gha) adjusts physical area to world-average biological productivity for the relevant account year. The NFA reports annual Ecological Footprint and biocapacity values in gha, with the annual basis implicit in the reported year (Global Footprint Network, 2009, 2019). The model uses the same annual account convention for B, E, Y_f, Y_c, and shortfall S; its land contributions are model quantities, not separate NFA series. The debt state D integrates the positive shortfall with leakage and is measured in gha yr, so $\dot{D}$ and ηD are in gha. Account values are not physical-stock measurements.

Table S1: Symbols, definitions, units, and roles. Per-hectare yields and contributions use the annual account basis (gha ha^{-1}). The coefficient $b_{G,c}$ is in gha yr ha^{-1}; annual account totals are in gha, with the accounting year implicit. The leaky debt state D is in gha yr.

Symbol	Meaning	Unit	Role
A_f	fast provisioning land (cropland, intensive)	ha	state
A_c	ecological-capital land (forest, soil, wetland, long-rotation systems; effective class)	ha	state
A_r	reserve land, $A_{tot} - A_f - A_c$	ha	dependent balance
q	per-hectare capital-land quality index	dimensionless	state
P	human population	cap	state

Symbol	Meaning	Unit	Role
D	leaky-debt state driven by aggregate deficit $E - B$	gha yr	state
b_f	fast-land account yield per hectare, $(b_{f0} + T_b)e^{-\alpha D}$	gha ha^{-1}	derived
$\tilde{b}_f(t)$	OWID/FAOSTAT cereal yield, a physical productivity sentinel (not all crops)	tonnes ha^{-1}	empirical input
$\hat{b}_f(t)$	Base-anchored index-scaled proxy yield used only in SI S5	gha ha^{-1}	proxy construction
b_{f0}	base fast-land yield per hectare	gha ha^{-1}	parameter
b_c	capital-land direct yield	gha ha^{-1}	parameter
$b_{G,c}$	capital regeneration-value coefficient	gha yr ha^{-1}	parameter
$G_c(q)$	per-ha regeneration of quality, $\rho_c q(1 - q/q_{\max})$	yr^{-1}	rate function
$\Upsilon_c(q)$	per-ha capital value, $b_c + b_{G,c}G_c(q)$ (also written $b_{c,\text{eff}}$)	gha ha^{-1}	derived
Δb_{conv}	composition diagnostic, $b_f - \Upsilon_c(q)$	gha ha^{-1}	derived
$Y_f,\ Y_c$	fast and capital yields, $b_f A_f$ and $\Upsilon_c A_c$	gha	derived
B	model biocapacity $Y_f + Y_c$; or NFA total under its account convention	gha	aggregate
E_{prod}	NFA Global Ecological Footprint of Production	gha	empirical account
E	model-defined footprint demand, eP (or exogenous frozen demand)	gha	aggregate
$X,\ C$	crop-proxy account $X = \hat{b}_f A_f$ and residual $C = B - X$; neither is physical capital	gha	illustrative accounting split
S	shortfall, $[E - \sigma_f Y_f - \sigma_c Y_c]_+$	gha	derived
u_c	conversion rate, $\min(S/b_f,\ A_c - A_c^{\min})/\tau_{\text{conv}}$	ha yr^{-1}	flow
K	carrying capacity, $\max(B/e, K_{\min})$	cap	algebraic
e	per-capita footprint	gha cap^{-1}	parameter
r	intrinsic population growth rate	yr^{-1}	parameter
ρ_c	quality regeneration rate	yr^{-1}	parameter
$q_{\max}$	quality ceiling	dimensionless	parameter
μ	cropland maintenance rate (reserve→fast)	yr^{-1}	parameter
η_f	fast-land retirement rate (fast→reserve)	yr^{-1}	parameter
τ_{conv}	conversion time constant	yr	parameter
α	degradation rate of b_f per unit debt	gha^{-1} yr^{-1}	parameter
η	debt-repayment rate	yr^{-1}	parameter
$\tau_g,\ \tau_p$	regeneration and demographic delays	yr	parameter
$\sigma_f,\ \sigma_c$	human-available shares of the two flows	dimensionless	parameter
R_{rc}	surplus-gated restoration, $\chi_r A_r\,[B - E]_+$ (reserve→capital)	ha yr^{-1}	gated flow
χ_r	restoration coefficient	gha^{-1} yr^{-1}	parameter
T_b	technology yield increment, $\omega_{Tb}\Delta b/(1 + e^{-\kappa_w(t - t_{wave})})$	gha ha^{-1}	off except in SI S8

Symbol	Meaning	Unit	Role
ω_{Tb}	activation flag for the technology wave	dimensionless	0 baseline; 1 in SI S8
Δb, κ_w, t_{wave}	amplitude, steepness and center of the wave	gha ha^{-1}; yr^{-1}; yr	active only in SI S8
$A_c^{\min}$, $A_r^{\min}$, $A_f^{\max}$, $K_{\min}$	capital viability floor, reserve floor, arable ceiling, carrying-capacity floor	ha, ha, ha, cap	typed floors
A_{tot}	total bioproductive area	ha	parameter
k	$\mu/(\mu+\eta_f)$, fast share of non-capital land at equilibrium	dimensionless	derived
$B_{eq}(A_c)$	frozen-demand balance branch, $b_{f0}kA_{tot}-(b_{f0}k-b_c)A_c$	gha	derived
E_{ceil}	demand ceiling, $B_{eq}(A_c^{\min})$	gha	derived
$\lambda_{1,2}$	deficit-side Jacobian eigenvalues	yr^{-1}	derived
R_B, R_A, ψ_f	E/B, E/Y_f, Y_f/B (so $R_A = R_B/\psi_f$)	dimensionless	diagnostics
ϕ_0, w_X, w_C	base-year cropland share and log-mean weights	dimensionless	empirical decomposition
$s_X, \bar{s}_X, \bar{s}_C$	current and average fast-land shares	dimensionless	empirical decomposition
$\mathcal{R}_c(T)$	recovery metric, $(A_c(T)-A_c^{\deg})/(A_c^{\text{init}}-A_c^{\deg})$	dimensionless	derived
R_{fc}, χ	fast→capital restoration variant, $\chi A_f\,[B-E]_+$	ha yr^{-1}; gha^{-1} yr^{-1}	extension (off)
L_c, H_c	timber liquidation flow $[H_c-\sigma_c Y_c]_+/b_{G,c}$ and its target	ha yr^{-1}; gha	extension (off)

Model extensions. The reported simulations do not include timber liquidation or direct restoration from fast land to capital land. SI §S6 considers reservations of the human-available shares as a policy extension. The technology wave is off in the baseline and gate-sign comparison; it is switched on only for the matched experiment in SI §S8.

Conventions. We apply the positive-part and minimum rules exactly. Conversion and restoration transfer area among A_f, A_c, and A_r; the ledger therefore conserves A_{tot}. Quality q changes per-hectare value, not land area. Reserve-to-fast maintenance can balance fast-land retirement at equilibrium. We enforce $A_c \geq A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$, $P, D \geq 0$, $K \geq K_{\min}$, and the reserve and arable bounds. All reported paths remain within those bounds; reserve area stays at least $0.80\,ha$ in the drawdown sweeps.

S2 Derivations and proofs

S2.1 The composition identity with varying debt and quality

Differentiate $B = b_f A_f + \Upsilon_c(q)A_c$, use $b_f = (b_{f0}+T_b)e^{-\alpha D}$, and substitute Eqs. (6)–(7) to obtain the general identity

$$\dot{B} = \Delta b_{\text{conv}}\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c}G_c'(q)A_c\,\dot{q} + \Upsilon_c R_{rc} - \alpha\dot{D}\,b_f A_f + \frac{\dot{T}_b}{b_{f0}+T_b}\,b_f A_f. \quad \text{(S1)}$$

The conversion term is positive exactly when $\Delta b_{\text{conv}} > 0$. Because $\dot{q} \geq 0$, the quality term is non-negative below $q_{\max}/2$ and non-positive above it. Debt accumulation reduces B; a rising

technology increment and reserve-to-capital restoration add to it. The main-text reduced comparison suppresses technology and restoration, fixes debt and quality, and balances maintenance with retirement.

In the separate demonstration run, debt is allowed to change: D first rises to 0.20 gha yr and then declines. At the start, conversion contributes $+0.21\,\mathrm{gha\,yr^{-1}}$ to $\dot{B}$; maintenance and retirement contribute $-0.005\,\mathrm{gha\,yr^{-1}}$, and debt contributes about $-0.004\,\mathrm{gha\,yr^{-1}}$. Under those initial conditions, conversion is the largest term in the change in total model biocapacity.

S2.2 The deficit-side affine system and its spectrum

On the deficit side, set $q = q_{\max}$, freeze D, switch off R_{rc}, and assume $S/b_f < A_c - A_c^{\min}$. Equation (5) then gives $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\mathrm{conv}})$. Using $A_r = A_{tot} - A_f - A_c$, Eqs. (6)–(7) reduce to

$$\frac{d}{dt}\begin{pmatrix} A_f \\ A_c \end{pmatrix} = \begin{pmatrix} -\tau_{\mathrm{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\mathrm{conv}}} \\ \tau_{\mathrm{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\mathrm{conv}}} \end{pmatrix} \begin{pmatrix} A_f \\ A_c \end{pmatrix} + \begin{pmatrix} \dfrac{E}{b_f \tau_{\mathrm{conv}}} + \mu A_{tot} \\ -\dfrac{E}{b_f \tau_{\mathrm{conv}}} \end{pmatrix}. \tag{S2}$$

The matrix has trace $\operatorname{tr} J = -\tau_{\mathrm{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f$ and determinant $\det J = (\mu + \eta_f)(k - b_c/b_f)/\tau_{\mathrm{conv}}$, where $k = \mu/(\mu + \eta_f)$. At baseline, $\operatorname{tr} J = -1.0512\,\mathrm{yr^{-1}}$ and $\det J = 0.05353\,\mathrm{yr^{-2}}$. Its eigenvalues are

$$\lambda_{1,2} = \tfrac{1}{2}\Big(\operatorname{tr} J \pm \sqrt{\operatorname{tr}^2 J - 4 \det J}\Big) = -0.0537\,\mathrm{yr^{-1}},\ -0.9975\,\mathrm{yr^{-1}}. \tag{S3}$$

Because J depends on neither demand E nor the current state, this spectrum stays constant along the frozen-demand branch. At $E/E_{\mathrm{ceil}} = 0.50$, 0.90, 0.99, 0.999, the slow eigenvalue remains $\lambda_1 = -0.05366\,\mathrm{yr^{-1}}$. It is unchanged as ρ_c varies ($\lambda_1 = -0.0537\,\mathrm{yr^{-1}}$ at $\rho_c = 0.04$, 0.08, $0.16\,\mathrm{yr^{-1}}$), shifts modestly with τ_{conv} (-0.0553, -0.0537, -0.0505, $-0.0417\,\mathrm{yr^{-1}}$ at $\tau_{\mathrm{conv}} = 0.5$, 1, 2, 5 yr), and shifts more with μ (-0.0254, -0.0537, $-0.1099\,\mathrm{yr^{-1}}$ at $\mu = 0.03$, 0.06, $0.12\,\mathrm{yr^{-1}}$). It remains negative in every tested sweep.

The affine system's fixed point is $A_c^*(E)$ from Eq. (18), with $A_f^* = k(A_{tot} - A_c^*)$. For $E > E_{\mathrm{ceil}}$, this point lies below $A_c^{\min}$. The trajectory instead approaches the boundary $A_c = A_c^{\min}$ asymptotically, with $u_c \to 0$ and persistent shortfall $S^* = E - E_{\mathrm{ceil}}$. At baseline, two members of the no-conversion equilibrium family are $(A_f^*, A_c^*, A_r^*, P^*) = (0.914\,\mathrm{ha}, 0.824\,\mathrm{ha}, 0.762\,\mathrm{ha}, 1.488\,\mathrm{cap})$ and $(1.091\,\mathrm{ha}, 0.500\,\mathrm{ha}, 0.909\,\mathrm{ha}, 1.731\,\mathrm{cap})$. Both satisfy $A_r^*/A_f^* = \eta_f/\mu = 0.8333$; the second is a sustainable cropland equilibrium supported by maintenance. This reserve-share mode explains why the strict-inequality interval in the delayed-quality demonstration lasts about 149 yr, even though most conversion is complete within 25 yr.

When the other branch of the minimum applies ($S/b_f \geq A_c - A_c^{\min}$), $u_c = (A_c - A_c^{\min})/\tau_{\mathrm{conv}}$. The distance to the floor then follows $\frac{d}{dt}(A_c - A_c^{\min}) = -(A_c - A_c^{\min})/\tau_{\mathrm{conv}}$: it shrinks exponentially but never reaches zero in finite time. The floor is approached asymptotically, so operational proximity thresholds depend on the tolerance used to define "near" (Section S4.2).

S2.3 Characteristic structure of the full delay system

Let $z = (A_f, A_c, q, P, D)$. On any fixed active set, the linearized delay system is $\dot{z}(t) = J_0 z(t) + J_g z(t - \tau_g) + J_p z(t - \tau_p)$, with characteristic equation $\det \Delta(\lambda) = 0$ and $\Delta(\lambda) = \lambda I - J_0 - J_g e^{-\lambda \tau_g} - J_p e^{-\lambda \tau_p}$. The current-state matrix J_0 contains the conversion and debt feedbacks: the 2×2 land block above, the deficit-side debt row $\partial \dot{D}/\partial z = (-b_f, -\Upsilon_c, -b_{G,c} G_c' A_c, e, -\eta)$, and the couplings $\partial b_f/\partial D = -\alpha b_f$. The regeneration-delay matrix J_g has the single entry $G_c'(q^*)$ in the q row. The population-delay matrix J_p has the row $(rP^{*2}/K^{*2})\, e^{-1}\, \partial B/\partial z$ in the P position, from $\partial \dot{P}/\partial K = rP^2/K^2$ and $K = B/e$.

At baseline, two scalar delayed modes decouple. Linearized at $q_{\max}$, the quality mode is $\dot{\xi} = -\rho_c \xi(t - \tau_g)$, with undelayed rate $-0.08\,\mathrm{yr}^{-1}$ and characteristic equation $\lambda = -\rho_c e^{-\lambda\tau_g}$. Since $\rho_c\tau_g = 1.6 > \pi/2$, this linearized mode would oscillate. The nonlinear model cannot follow that oscillation: $\dot{q} = G_c(q(t-\tau_g)) \geq 0$ for $q(t-\tau_g) \in [0, q_{\max}]$, and q is clamped at $q_{\max}$. Quality is therefore non-decreasing and converges to $q_{\max}$; the oscillation is an artifact of linearizing a one-sided law at its boundary. On the no-conversion face, the population mode follows the delayed logistic law with $r\tau_p = 0.5 < \pi/2$, below the Hutchinson threshold, so it is non-oscillatory at baseline.

S2.4 Debt fixed points beyond the ceiling

For $E > E_{\mathrm{ceil}}$, capital-land area approaches $A_c = A_c^{\min}$ and the persistent shortfall feeds debt. At the floor, $B_{eq}(A_c^{\min}; D) = b_{f0}e^{-\alpha D}k(A_{tot} - A_c^{\min}) + b_c A_c^{\min}$, so $\dot{D} = E - B_{eq}(A_c^{\min}; D) - \eta D$. Its fixed point $D^*(E)$ satisfies $\eta D = E - B_{eq}(A_c^{\min}; D)$. The right-hand side is increasing in D, making the fixed point unique and attracting when $\eta > \alpha b_{f0} k(A_{tot} - A_c^{\min})$; at baseline, $0.034\,\mathrm{yr}^{-1} < 0.05\,\mathrm{yr}^{-1}$. Baseline values are $D^* = 1.186\,\mathrm{gha\,yr}$ at $E = 1.158\,\mathrm{gha}$ and $D^* = 5.375\,\mathrm{gha\,yr}$ at $E = 1.238\,\mathrm{gha}$. Corresponding $B(D^*)$ values are 1.099 and 0.969 gha, with persistent shortfalls $S^* = 0.059$ and 0.269 gha. Under fixed demand, debt therefore saturates; it is unbounded only if the shortfall is unbounded.

S2.5 Boundedness asymmetry and the crossover shortfall

Equation (12) gives $\ln b_f = \ln(b_{f0} + T_b) - \alpha D$. The technology term is bounded between $\ln b_{f0}$ and $\ln(b_{f0} + \Delta b)$, so its maximum contribution to $\ln b_f$ is $\ln(1 + \Delta b/b_{f0}) = \ln(1.85/0.85) = 0.778$ at baseline. From Eq. (13), $\dot{D} \leq \bar{S} - \eta D$, where $\bar{S}$ is the running supremum of shortfall. Thus $D(t) \leq \max\{D(0), \bar{S}/\eta\}$, and the debt channel contributes no less than $-\alpha\bar{S}/\eta$. The channels are equal in magnitude at $\bar{S} = S^\dagger = (\eta/\alpha)\ln(1 + \Delta b/b_{f0}) = 1.296\,\mathrm{gha}$.

The fast-land contribution $Y_f = b_f A_f$ has the same two-channel structure; capital contribution $\Upsilon_c A_c$ is unaffected by either. Under the additive law $b_f = b_{f0} + T_b - \alpha D$, the effects are directly comparable as bounded increments (Δb versus αD^*), so the logarithmic asymmetry disappears. The bound above applies to the multiplicative specification.

S2.6 Macro-ratio identities

Set $R_B = E/B$, $R_A = E/Y_f$, and $\psi_f = Y_f/B$. Then $R_A = R_B/\psi_f$ by definition. The direct-yield share of capital contribution, $\psi_c = b_c A_c/Y_c$, is a different ratio; $R_A = R_B/\psi_c$ is not an identity. Aggregate balance occurs at $R_B = 1$. With baseline shares $\sigma_f = \sigma_c = 1$, $R_A > 1$ is necessary but not sufficient for conversion: shortfall requires $E > B = Y_f + Y_c$, not merely $E > Y_f$. The threshold changes when flows are reserved. Capital-land area falls when $u_c > R_{rc}$, with u_c defined in Eq. (5). Thus, under baseline shares, conversion requires $R_B > 1$. Under the reservation extension with $\sigma_f, \sigma_c < 1$, conversion and declining A_c can occur even when $R_B \leq 1$; aggregate overshoot is not required. At $q^* = q_{\max}/2$, the direct-yield share of capital value is $b_c/(b_c + b_{G,c}\rho_c q_{\max}/4) = 0.758$ at baseline. This value applies at that point; it is not a structural constant.

S3 Numerical methods and sensitivity analyses

Numerical method. We integrate the delay equations by the method of steps with explicit first-order Euler updates. Pre-run histories of B and q are constant at their initial values over $[-\max(\tau_g, \tau_p), 0)$. In the technology-wave comparison, both runs share the same no-wave history; the active logistic tail changes $B(0)$, and subsequent wave-adjusted values enter the delayed carrying-capacity input $K(t - \tau_p)$. Area balance is enforced by calculating $A_r = A_{tot} - A_f - A_c$.

We apply the positive-part operator exactly and keep $A_c \geq A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$, $P, D \geq 0$, and $K \geq K_{\min}$.

Resolution and threshold reporting. The demonstration and threshold bisection use $\Delta t = 0.05\,\text{yr}$; broad drawdown sweeps use 0.1 yr, with selected thresholds checked on the finer step. At $A_{c0} = 1.0\,\text{ha}$, the 10^{-6} ha floor-proximity value is 2.955 cap on the sweep resolution and 2.98 cap under the finer bisection. We report these as tolerance- and resolution-dependent proximity values, not as a discontinuous critical population.

Delayed versus current quality. Reported trajectories use the delayed regeneration term $G_c(q(t-\tau_g))$. We also test the current-quality alternative $G_c(q(t))$. The current-quality version extends the demonstration interval by 1.3 yr and shifts the tested 10^{-6} ha population proximity value by at most 0.001 cap (Table S2). These differences are small in this comparison; they do not establish that delay is unimportant outside the tested range.

Table S2: Sensitivity to evaluating regeneration at current rather than delayed quality. The baseline parameter set and $\tau_p = 25\,\text{yr}$ are used; the reported trajectories use the delayed-quality specification. Biocapacity B, fast-land contribution Y_f, and capital-land contribution $\Upsilon_c(q)A_c$ are in gha; land areas A_c, A_f, A_{c0} are in ha; the operational initial-population floor-proximity threshold P_0 is in cap; delays and step intervals are in yr.

Quantity	current $q(t)$	delayed $q(t-\tau_g)$
Demonstration window with $\dot{B} > 0$, $\dot{A}_c < 0$ (yr)	150.3	149.0
Mask-end values B, A_c, A_f	0.825, 0.808, 0.923	0.825, 0.808, 0.923
Capital contribution $\Upsilon_c(q)A_c$ (initial $\to$ mask end, gha)	0.08364 $\to$ 0.04038	0.08364 $\to$ 0.04038
Fast-land contribution Y_f (initial $\to$ mask end, gha)	0.51000 $\to$ 0.78462	0.51000 $\to$ 0.78462
P_0 floor threshold ($A_{c0} = 1.0\,\text{ha}$, $\tau_g = 5\,\text{yr}$, $\Delta t = 0.1\,\text{yr}$)	2.955	2.955
P_0 floor threshold ($A_{c0} = 1.0\,\text{ha}$, $\tau_g = 20\,\text{yr}$)	2.955	2.954
P_0 floor threshold ($A_{c0} = 1.0\,\text{ha}$, $\tau_g = 40\,\text{yr}$)	2.955	2.954

Well-posedness. The bounds define a forward-invariant state space $\Omega = \{(A_f, A_c, A_r, q, P, D) : A_i \geq 0,\ q \in [0, q_{\max}],\ A_f + A_c + A_r = A_{tot},\ K \geq K_{\min}\}$. The positive-part and minimum operators are Lipschitz, so each admissible history yields a unique delay-system solution. At the switching surface $S = 0$, stability is evaluated from the deficit side.

S4 Extended model results

S4.1 Fast-land yield sweep and parameter sensitivity of the composition diagnostic

Table S3 varies fast-land yield while holding the other baseline parameters fixed. It reports the composition diagnostic, the deficit-side eigenvalues, the balance-branch slope, and the demand ceiling. The diagnostic stays positive across the four tested yields.

A separate sensitivity check varies the regeneration-value coefficient $b_{G,c}$. At $b_f = 0.85\,\text{gha ha}^{-1}$, increasing $b_{G,c}$ from 0.4 to $1.2\,\text{gha yr ha}^{-1}$ raises $\Upsilon_c(q^*)$ from 0.058 to $0.074\,\text{gha ha}^{-1}$; the corresponding conversion diagnostic remains positive, declining from +0.792 to $+0.776\,\text{gha ha}^{-1}$. These parameter sweeps show how the sign behaves over the tested ranges; they are not estimates of land-class yields.

Table S3: Fast-land yield sweep at the baseline (all other parameters fixed): composition diagnostic at q^*, eigenvalues of the deficit-side matrix, slope of the balance branch and demand ceiling.

b_f (gha ha^{-1})	$\Delta b_{\mathrm{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	$b_f k - b_c$ (gha ha^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.277	0.804
0.85	+0.784	−0.0537	−0.998	0.414	1.138
1.20	+1.134	−0.0547	−1.014	0.605	1.606
2.00	+1.934	−0.0556	−1.029	1.041	2.675

S4.2 Drawdown-depth ladder

Table S4 records the lowest capital-land area in each endogenous-demand run. The grid varies initial population P_0 and initial capital area A_{c0}; all other settings are listed in the table note. The minimum occurs 3–25 yr after the run starts, when the shortfall first closes ($B \geq E$). The full P_0 grid (step 0.1) is also preserved in `si_drawdown_depth.csv`.

The proximity thresholds are tolerance-dependent because A_c approaches the floor asymptotically. For $A_{c0} = 0.40$ ha, the populations at which the trajectory comes within 10^{-2}, 10^{-3}, and 10^{-4} ha of the floor are 1.8, 1.9, and 2.1 cap. The corresponding triples are 2.2, 2.3, and 2.4 cap at $A_{c0} = 0.70$ ha; 2.6, 2.7, and 2.8 cap at 1.00 ha; and 2.8, 2.9, and 3.0 cap at 1.18 ha. A finer bisection using a 10^{-6} ha criterion, $\Delta t = 0.05$ yr, and $T = 1000$ yr gives 2.41, 2.70, 2.98, and 3.14 cap, respectively. Switching χ_r from 0.10 to 0 leaves the bisection thresholds unchanged to three decimals: 2.387 cap for $A_{c0} = 0.40$ ha and 2.955 cap for $A_{c0} = 1.00$ ha ($\Delta t = 0.1$ yr). Across the tested grid, drawdown depth varies continuously and monotonically with both P_0 and A_{c0}.

At $A_{c0} = 1.00$ ha, varying the regeneration delay over $\tau_g = 5, 20, 40$ yr changes the delayed-quality 10^{-6} ha proximity value by at most 0.001 cap (Table S2). This comparison covers only the stated initial conditions and delay range.

Table S4: Minimum capital land $\min_t A_c$ (ha) reached during the run, by initial capital A_{c0} (rows, ha) and initial population P_0 (columns, cap); typed floor $A_c^{\min} = 0.05$ ha.

$A_{c0} \setminus P_0$	1.0	1.5	2.0	2.5	2.8	3.0	3.5	4.0
0.4	0.396	0.147	0.050	0.050	0.050	0.050	0.050	0.050
0.7	0.700	0.442	0.121	0.050	0.050	0.050	0.050	0.050
1.0	1.000	0.731	0.398	0.075	0.050	0.050	0.050	0.050
1.18	1.180	0.898	0.557	0.230	0.055	0.050	0.050	0.050

S4.3 Recovery metric and the role of the restoration flow

We summarize recovery with $\mathcal{R}_c(T) = (A_c(T) - A_c^{\text{deg}})/(A_c^{\text{init}} - A_c^{\text{deg}})$. A value of 0 means no recovery, 1 means return to the reference, values below 0 indicate further decline, and values above 1 indicate over-restoration.

Table S5 applies the metric to delayed-quality runs with $A_{c0} = 1.0$ ha and $T = 400$ yr. Without restoration, there is no return flow and A_c remains at its minimum. Surplus-gated restoration produces partial recovery in all six initial-population cases ($\mathcal{R}_c = 0.21$–0.89); none reaches the reference by year 400. The gate-sign experiment in Section S4.4 uses a different post-drawdown state and includes interventions that over-restore.

Table S5: Recovery of the capital-land area after a demand-driven drawdown ($A_{c0} = 1.0$ ha, $A_{f0} = 0.6$ ha, $q_0 = 0.9$, $T = 400$ yr): initial population P_0 is in cap, initial footprint E_0 in gha, capital-land values in ha, and the recovery metric is dimensionless.

P_0	E_0	$\min_t A_c$	t(min) (yr)	$A_c(T)$, $\chi_r = 0$	$A_c(T)$, $\chi_r = 0.10$	$\mathcal{R}_c(T)$, $\chi_r = 0.10$
1.5	0.825	0.731	3.0	0.731	0.969	0.89
2.0	1.100	0.398	3.8	0.398	0.677	0.46
2.5	1.375	0.075	4.1	0.075	0.433	0.39
3.0	1.650	0.050	13.8	0.050	0.328	0.29
3.5	1.925	0.050	20.6	0.050	0.278	0.24
4.0	2.200	0.050	26.6	0.050	0.250	0.21

S4.4 Gate-sign phase dependence

The gate-sign comparison starts from a post-drawdown state: $A_c = 0.058$ ha (near the floor), $A_f = 0.6$ ha, $q = 0.9$, and $P = 1.2$ cap. At the degraded equilibrium, $R_B = 1$ and $B = E$, so neither restoration gate is active. The reference equilibrium has $A_c^{\text{init}} = 0.8225$ ha and $P^{\text{init}} = 1.489$ cap; the degraded reference is $A_c^{\text{deg}} = 0.0513$ ha. Every row uses $T_b = 0$ and the same starting state.

Table S6: Restoration gate and recovery outcomes at $T = 100$ and 900 yr. The table reports end-state land areas, population relative to the reference, trajectory extrema, and peak debt D (gha yr). Annual account values $B = E$ are in gha; land areas are in ha; population ratios and recovery metrics are dimensionless. Each row approaches $R_B = 1$ ($E = B$).

Policy / gate	$\mathcal{R}_c(100)$	$\mathcal{R}_c(900)$	A_c	A_f	A_r	$B = E$	P/P^{init}	$\min A_r$	$\max A_c$	D_{peak}
none ($R_{rc} = 0$)	−0.00	−0.00	0.051	1.336	1.113	1.138	1.39	1.113	0.057	0.152
surplus-gated, $\chi_r = 0.02$	0.69	0.73	0.612	1.030	0.858	0.906	1.11	0.858	0.612	0.152
surplus-gated, $\chi_r = 0.05$	1.08	1.07	0.875	0.886	0.738	0.797	0.97	0.724	0.885	0.152
surplus-gated, $\chi_r = 0.10$	1.28	1.27	1.030	0.802	0.668	0.733	0.90	0.602	1.092	0.152
surplus, $\chi_r = 0.10$, target cap $A_c^* = 0.8225$	1.00	1.00	0.822	0.915	0.763	0.819	1.00	0.664	0.822	0.152
surplus, $\chi_r = 0.10$, conversion frozen	1.35	1.36	1.097	0.765	0.638	0.705	0.86	0.602	1.097	0.543
deficit-gated, $\chi_r = 0.05$	0.00	0.00	0.055	1.334	1.112	1.136	1.39	1.112	0.060	0.145
deficit-gated, $\chi_r = 0.10$	0.01	0.01	0.059	1.331	1.110	1.135	1.39	1.110	0.065	0.139
deficit, $\chi_r = 0.10$, conversion frozen	0.05	0.05	0.088	1.315	1.096	1.123	1.37	1.096	0.088	0.159

Without restoration, A_c remains at its minimum because the model has no return flow. Under surplus-gated restoration, outcomes depend on the restoration rate: the lowest rate yields partial recovery, while higher rates can exceed the reference. At $\chi_r \geq 0.05$, restoration continues while $B > E$ and the supported population ends at 0.97–0.90 of its reference level. A target cap returns A_c to its reference ($\mathcal{R}_c = 1.00$).

Deficit-gated restoration acts during shortfall. At the tested rates, conversion is stronger and recovery stays near zero. Freezing conversion under surplus-gated restoration raises peak debt from 0.152 to 0.543 gha yr; restoration shifts reserve area toward capital land while fast-land maintenance declines. These outcomes depend on this starting state and the selected intervention rates. The gate-sign comparison sets $T_b = 0$; Section S8 reports the separate matched technology-wave/debt experiment.

S4.5 Regime table

Table S7: The composition diagnostic Δb_{conv} (gha ha^{-1}) as the decision condition for mask visibility. There is no interior area-based capacity maximum in the quality-based model; the only shape distinction is the sign of Δb_{conv}.

Regime	Condition	Behavior
composition-premium	$\Delta b_{\text{conv}} > 0$	the conversion contribution to accounting B is positive; net B also depends on other flows (the mask is possible in the baseline)
composition-neutrality	$\Delta b_{\text{conv}} = 0$	first-order neutrality; non-generic
composition-discount	$\Delta b_{\text{conv}} < 0$	conversion lowers B at the conversion margin; conversion alone cannot produce a composition mask

S5 Empirical details

S5.1 Data and vintages

Our empirical source is the National Ecological Footprint and Biocapacity Accounts (NFA), Edition 2026, results and metadata version 1.0 (Lo et al., 2026). The workbook spans 1961–2025; we use complete world observations through 2023 and exclude the estimated 2024–2025 rows. The 1961–2022 sensitivity uses the same edition and population series. This release recalculates historical values and reports five biocapacity account categories: fishing grounds, built-up land, cropland, grazing land, and forest biocapacity. These are account categories, not a common-boundary inventory of physical stocks.

The workbook field `pop_1000_persons` reports UN PopStat population in thousands; we convert it to persons before calculating per-capita values. The field `efp_total_gha` is the Global Ecological Footprint of Production, denoted E_{prod}. Per-capita quantities are recalculated from unrounded account totals and population, rather than copied from rounded workbook columns.

Table S8 summarizes the endpoints. From 1961 to 2023, total biocapacity account value B rises from 9,725,417,562 to 12,142,843,961 gha, while population grows from 3.06487 to 8.09173 billion. Over the same period, B/P falls from 3.17319 to 1.50065 gha person^{-1} and E_{prod}/P rises from 2.19700 to 2.56462 gha person^{-1}. The accounting identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$ holds to rounding; it separates extensive from per-person change but does not attribute causes. In both windows, 1973 is the first year with $E_{\text{prod}}/B > 1$.

Table S8: World endpoint changes in NFA Edition 2026. The 1961–2022 column is a same-edition sensitivity; 2024–2025 estimates are excluded. Population and per-capita quantities are recomputed from workbook totals.

Measure	1961–2023	1961–2022
$\Delta B/B_{1961}$	+24.8568%	+24.5450%
$\Delta \ln B$	+0.221997	+0.219497
$\Delta P/P_{1961}$	+164.0154%	+161.7211%
$\Delta \ln P$	+0.970837	+0.962109
$\Delta(B/P)/(B/P)_{1961}$	−52.7085%	−52.4131%
$\Delta \ln(B/P)$	−0.748840	−0.742612
$\Delta(E_{\text{prod}}/P)/(E_{\text{prod}}/P)_{1961}$	+16.7327%	+17.6994%
First year $E_{\text{prod}}/B > 1$	1973	1973

S5.2 Divisia decomposition, sensitivity to the base-year share and level fallback

The crop proxy is an illustrative account construction. We combine independently sourced OWID/FAOSTAT world cropland area, defined as arable plus permanent-crop area (ha), with cereal yield (tonnes per hectare) (Our World in Data, 2026a,b). Cereal yield is a productivity sentinel, not a measure of output from all crops. For a declared 1961 account-share anchor ϕ_0, we set $\hat{b}_f(t) = \hat{b}_f(1961)\tilde{b}_f(t)/\tilde{b}_f(1961)$, with $\hat{b}_f(1961)A_f(1961) = \phi_0 B(1961)$. We then define $X_t = \hat{b}_f(t)A_f(t)$ and $C_t = B_t - X_t$. Thus X is an index-scaled account proxy, and C is the residual that reconciles it to the NFA total. The residual is not observed ecological capital, non-cropland biocapacity, or the model quantity $\Upsilon_c(q)A_c$.

When both X and C are positive at the two endpoints, the exact log-mean Divisia identity is

$$\Delta \ln B = w_X\left(\Delta \ln \tilde{b}_f + \Delta \ln A_f\right) + w_C \Delta \ln C, \qquad w_X = \frac{L(X_T, X_0)}{L(B_T, B_0)}, \quad w_C = \frac{L(C_T, C_0)}{L(B_T, B_0)}, \tag{S4}$$

Here $L(a, b) = (b - a)/(\ln b - \ln a)$ is the logarithmic mean, with its continuous limit when $a = b$. The central anchor is the measured 1961 NFA cropland account share, $\phi_0 = 0.1286472202$. Mapping that share onto the external area–yield proxy is an assumption; it is not an independent observation of physical capital.

At this anchor, the 1961–2023 cereal-yield, area, and residual contributions are +0.268662, +0.037710, and −0.084375, summing to $\Delta \ln B = +0.221997$. For 1961–2022, the corresponding values are +0.264696, +0.037506, and −0.082705, summing to +0.219497. Table S9 shows how the allocation changes with ϕ_0. The identity is exact for the stated proxy; the contribution assigned to each term depends on the anchor and is not a physical-stock attribution.

Table S9: Sensitivity of the proxy Divisia decomposition to the 1961 account-share anchor ϕ_0 (world, 1961–2023). Contributions and weights are dimensionless. The $\phi_0 = 0.128647$ row uses the measured NFA cropland share in 1961; other rows are illustrative calibrations.

ϕ_0	w_X	w_C	Cereal yield	Area	Residual C	Sum
0.100000	0.1833	0.7957	+0.2088	+0.0293	−0.0162	+0.2220
0.128647	0.2358	0.7352	+0.2687	+0.0377	−0.0844	+0.2220
0.150000	0.2749	0.6893	+0.3133	+0.0440	−0.1352	+0.2220
0.190000	0.3483	0.6008	+0.3968	+0.0557	−0.2305	+0.2220
0.250000	0.4582	0.4581	+0.5221	+0.0733	−0.3734	+0.2220
0.300000	0.5499	0.3178	+0.6265	+0.0879	−0.4924	+0.2220
0.320000	0.5866	0.2454	+0.6683	+0.0938	−0.5401	+0.2220
0.340000	0.6232	0.1009	+0.7100	+0.0997	−0.5877	+0.2220

For $\phi_0 \geq 0.340531$ in 2023 (or 0.343400 in 2022), terminal residual C is nonpositive, so logarithmic weighting is invalid. We use the exact level identity $\Delta B = \Delta X + \Delta C$ instead; this is still an accounting identity, not a stock measure. At the selected anchor, $X_{2023}/B_{2023} = 0.377784$ and C remains positive, but the proxy fast-land share and residual contribution vary across the sensitivity analysis. Every row uses the same NFA total B. The residual absorbs differences between the index-scaled crop proxy and that total, including account categories, boundaries, valuation, vintage, and calibration.

S5.3 Cropland-account calibration

NFA cropland-account movements are distinct from the illustrative crop proxy. From 1961 to 2023, NFA cropland biocapacity rises 2.870055-fold and NFA cropland area rises 1.198029-fold; their account-value-per-area quotient rises 2.395647-fold. The same-edition 1961–2022 ratios are 2.851441, 1.198326, and 2.379520. This quotient is an account value per area, not an independent physical-yield measure.

Over 1961–2023, the external OWID/FAOSTAT cropland-area series rises 1.173414-fold and cereal yield rises 3.124667-fold. These independent series serve only as a productivity sentinel and area input; together they do not reconstruct the NFA cropland account or measure output from all crops. Accordingly, the model's representative $b_{f0} = 0.85\,\mathrm{gha\,ha^{-1}}$ is not fitted to these ratios.

S5.4 Observation-operator hierarchy

- *NFA accounting observations.* Total and component values in gha describe NFA account composition and change under the stated edition and conventions. They do not directly measure biomass, soil condition, wetland extent, fish biomass, or model state A_c.
- *Fast-provisioning area.* The external world cropland-area series covers arable land plus permanent crops for 1961–2023 (ha) (Our World in Data, 2026b). It is an independent proxy input, not the NFA cropland-area series.
- *Crop-productivity sentinel.* OWID/FAOSTAT cereal yield is reported in tonnes per hectare for 1961–2024 (Our World in Data, 2026a). It covers cereals, not all crops, and is not a physical-stock measure. Base-anchored $\hat{b}_f$ is used only in the illustrative account proxy.
- *Ecological-capital components.* Separate physical observation operators could use forest extent and biomass, soil carbon and condition, wetland extent and condition, and fish biomass by species or stock. Such data would need harmonized boundaries, reference years, definitions, uncertainty, and explicit treatment of overlap. We assembled no common-boundary global physical-stock series for these components. NFA forest biocapacity and residual C cannot substitute for those measurements.
- *Model mapping.* The model defines capital contribution as $Y_c = \Upsilon_c(q)A_c$. Neither A_c nor q is identified empirically by total B or an NFA component, and we do not assume proxy residual $C = B - X$ equals Y_c. A model–data comparison would first need independent physical measures and an explicit mapping.

Table S10 summarizes the possible links while keeping NFA categories distinct from model states.

Table S10: Conceptual mapping from two-land model quantities to NFA information and independent proxy inputs. The rows specify possible observation operators, not identities between NFA categories and model states. Annual account values are in gha; land areas are in ha.

Model quantity	Closest account or proxy information	Interpretation and identification limit
Fast area and proxy account A_f, $X = \hat{b}_f A_f$	NFA cropland account; independently sourced OWID cropland area and cereal yield	NFA cropland biocapacity is an account value; the external area–yield product is an illustrative, base-anchored proxy. Neither is an observed model fast-land yield or a one-to-one physical measure.
Capital area, quality and contribution A_c, q, $Y_c = \Upsilon_c(q)A_c$	No direct NFA counterpart; the proxy residual $C = B - X$ is defined by subtraction	In the model $Y_c = \Upsilon_c(q)A_c$; empirically C is not assumed equal to Y_c. It also absorbs differences in boundaries, account categories (including built-up land), valuation, vintage and proxy calibration, and is not a physical-stock measure.
Reserve area A_r	No direct NFA account counterpart	A_r closes the model's conserved-area ledger; it is not recoverable from the NFA aggregate.

Table S11: Observability ladder: what is directly measured, what is identifiable by construction, and what remains unidentified.

Tier	Quantities	Status
Observable physical area input	OWID cropland area [ha]; other mapped land-cover classes where available	Measured extent, but mapping to effective A_c is not direct
Observable population	P [persons]	Direct in the UN PopStat workbook field after unit conversion
Observable productivity sentinel	Cereal yield [tonnes ha^{-1}]	Physical yield of cereals only; not all-crops output or ecological capital
Observable annual account value	NFA B, E_{prod} and components [gha]	Measured only under the NFA accounting convention; value-weighted accounts
Proxy account and residual	$X = \hat{b}_f A_f$ and $C = B - X$ [gha]	Identified by construction for the stated inputs and anchor; not physical stocks and not assumed equal to $\Upsilon_c(q)A_c$
Effective capital contribution	$Y_c = \Upsilon_c(q)A_c$ [gha]	A model quantity; no common-boundary physical observation operator is assembled here
Not separately identifiable	A_c, q, b_c, $b_{G,c}$, ρ_c and D	Requires independent physical-state, quality, regeneration or degradation observations and a declared mapping

S5.5 Regeneration anchors

The model's capital-land regeneration timescale is a lumped recruitment response, not a universal ecological rate. For Neotropical secondary forests, plot-level age and above-ground biomass records (2ndFOR; 1,334 plots, 41 chronosequences, 35 good fits) give median $t_{50} \approx 30$ yr and $\tau_r \approx 38$ yr under a lagged saturating recovery model (Poorter et al., 2016). For 33 fish stocks classified as collapsed in RAM Legacy, B/B_{MSY} trajectories give $t_{50} \approx 5.6$ yr and $\tau_r \approx 3.4$ yr (Hutchings and Reynolds, 2004; Neubauer et al., 2013). Soil organic carbon re-equilibrates over about 17–23 yr after land-use change (Poeplau et al., 2011). Separate forest evidence places biomass at about half of old-growth levels by 20 yr and at 90% by a median of about 66 yr; about 29% of collapsed fish stocks recover to 50% within 5–15 yr.

These observations describe different systems and are not pooled into one rate. Forest and soil records give decadal-scale context for the representative effective response scale of about 29 yr; the 10–40 yr interval is a modeling range for the lumped capital-land class, not a pooled estimate. The faster fish-stock episodes are not treated as a lower bound for that class.

In the quality law $G_c(q) = \rho_c q(1 - q/q_{\max})$, $1/\rho_c$ is a rate scale, not an empirically estimated recovery half-time. A half-time also depends on the starting state and the reference level. Neither ρ_c nor the delay τ_g is calibrated from these records.

S5.6 Recursive identifiability in the land-type accounts

NFA Edition 2026 illustrates why the aggregate should be read with its components, without treating account movements as physical-stock changes. From 1961 to 2023, cropland contributes +24.0577 percentage points to the +24.8568% change in total B; built-up land adds +7.4416, while Forest biocapacity contributes −5.9525, grazing land −0.2688, and fishing grounds −0.4212. The non-cropland sum, which includes built-up land, changes by only +0.7990 points (1.0092×) because unlike account movements offset one another.

For the same-edition 1961–2022 sensitivity, contributions are +23.8183 points for cropland, +7.2854 for built-up land, −5.8137 for Forest biocapacity, −0.2890 for grazing land, and −0.4559 for fishing grounds. The non-cropland sum changes by +0.7267 points (1.0083×). These are accounting attributions, not evidence that forest biomass, soil condition, wetland extent, fish biomass, or model A_c declined. Built-up land is an NFA account class, not ecological capital; including it in the non-cropland sum is arithmetic, not an ecological aggregation.

S5.7 Physical-stock scoreboard: specification, not inventory

No common-boundary global physical-stock series was assembled, so the scoreboard below is a specification of candidate measures, not an inventory, endpoint dataset, or model calibration. It is deliberately unpopulated.

The candidate quantities are heterogeneous: biomass and soil carbon are masses; wetland and cropland extent are areas; fish biomass is stock-specific; freshwater storage is a volume; and condition indicators may be dimensionless or categorical. They may also overlap spatially. Annual harvest, catch, increment, recharge, and emissions are flows or pressures, not stocks or extents. A composite would require common spatial and temporal boundaries, a declared normalization or valuation, uncertainty treatment, and explicit substitution rules. Until those choices are made, the components remain a nonadditive vector; no scalar composite follows from NFA gha accounts. SEEA Ecosystem Accounting likewise distinguishes extent and condition from ecosystem services and monetary presentations (United Nations et al., 2021).

The model's A_c is an effective area state and q a separate quality index; neither is identified by the NFA total. Physical observations could inform these states only through a declared mapping with compatible boundaries. The proxy residual $C = B - X = B - \hat{b}_f A_f$ is not assumed to equal $\Upsilon_c(q)A_c$.

Table S12: Physical-stock scoreboard specification. The rows are unpopulated: no common-boundary global physical-stock series for the relevant components or endpoint values were assembled. Stocks, extents, condition measures, and annual flows are not combined into a composite.

Component	Physical stock, extent, or condition to define	Annual flow or pressure (separate)	Comparability rule required
Forests	Extent (ha); growing stock or biomass (m^3 or tonnes C); stated condition	Net annual increment, harvest, or deforestation	Common forest definition and mapped area; reference year; treatment of plantations and overlap
Soils	Soil organic-carbon stock or density for a stated depth; structure or erosion condition	Carbon accumulation or loss; erosion where measured	Consistent depth, bulk density, land cover, and sampling method
Wetlands	Extent (ha) and an explicit ecological-condition measure	Carbon accumulation, water-storage change, or nutrient flux	Wetland definition and overlap with forest and other land-cover classes
Fisheries	Spawning-stock biomass by species or stock (tonnes); age or size structure	Recruitment, catch, and landings	Stock identity, assessment method, reference point, and uncertainty
Freshwater	Surface-water or groundwater storage (m^3), level, and quality	Recharge and withdrawal	Basin or aquifer boundary; separate stored volume from annual flow

S6 Policy extensions

Reserved biocapacity share. If both land classes have the same reservation share, $\sigma_f = \sigma_c = \sigma < 1$, human-available supply is σB and shortfall is $S = [E - \sigma B]_+$. Reservation lowers the demand at which conversion begins; conversion is off while $E \leq \sigma B$. With different shares, the condition becomes $E \leq \sigma_f Y_f + \sigma_c Y_c$, which need not equal σB. These results apply to the stated parameterization; they do not establish robustness to disturbances or parameter uncertainty.

Typed floors and conversion stop. The model imposes separate constraints on reserve land ($A_r \geq A_r^{\min}$), capital land ($A_c \geq A_c^{\min}$), fast land ($A_f \leq A_f^{\max}$), and carrying capacity ($K \geq K_{\min}$). A stop rule triggered at $A_c = A_c^{\min}$ preserves the capital-land floor. A ceiling on A_f limits fast-land extent but does not ensure that A_c stays above its floor unless the stop rule also accounts for the capital floor and remaining-area constraints. In Eq. (5), conversion slows to zero as $A_c \to A_c^{\min}$. A control based only on total B cannot guarantee this typed floor because B does not identify A_c.

Institutional signature. Related work distinguishes protective and mobilizing institutional feedback (Abaee, 2026a). Here, surplus-gated restoration is off during shortfall, while deficit-gated restoration acts during shortfall and can further reduce provision (Section S4.4).

Aggregation and typed feasibility. A related transition-assessment study defines a linear compensatory operator and compares perfect substitutability with less-substitutable CES alternatives (Sections 2 and 4 of Abaee, 2026b). ECOMOD's B is also a linear sum, but the companion considers a separate finite-plan setting. There, the compensatory protocol accepts states even when no available plan meets every component floor (Theorem 2 in Abaee (2026b)). Theorem 4(iii) in Abaee (2026b) identifies an exact interval of diagonal-gap states on its witness-defined region that each positive elasticity falsely certifies as safe. Thus no positive-elasticity member is uniformly safe on that region. Under pathwise feasibility, falling below a typed floor during transition remains a violation even if the endpoint later recovers. The comparison distinguishes an aggregate criterion from constraints on individual components.

S7 The one-stock comparator

We include this one-stock formulation as a structural comparator, not as a reduction of the two-land model. Its state A has comparator-specific stock units and is distinct from two-land state A_c. The equations are

$$
\begin{aligned}
&G(A) = \rho A\left(1 - \frac{A}{A_{\max}}\right), \qquad B(t) = b(t)A(t) + b_G G(A(t)), \qquad E(t) = eP(t);\\
&\dot{A}(t) = G\big(A(t-\tau_g)\big) - \frac{[E(t) - \sigma b(t)A(t)]_+}{b_G};\\
&\dot{P}(t) = rP(t)\left(1 - \frac{P(t)}{K(t-\tau_p)}\right), \qquad K(t) = \max\left(\frac{B(t)}{e}, K_{\min}\right);\\
&\dot{D}(t) = [E(t) - B(t)]_+ - \eta D(t), \qquad b(t) = \big(b_0 + T_b(t)\big)e^{-\alpha D(t)};\\
&T_b(t) = \omega_{Tb}\frac{\Delta b}{1 + \exp[-\kappa(t - t_w)]}.
\end{aligned}
$$

Here σ is the dimensionless human-available share; $\omega_{Tb} \in \{0, 1\}$ switches the logistic technology increment on or off. The states A, $A_{\max}$, and imposed floor A_{ext} use comparator-specific stock units. $A_{\max}$ sets the logistic growth scale; it is not an imposed upper bound, so $G(A)$ is in stock units per year. The coefficients b, b_0, T_b, and Δb are in gha per stock unit, while b_G is in gha yr per stock unit; both terms in B are annual account values in gha. Unlike the two-land transfer u_c, which divides shortfall by fast-land yield and applies a conversion timescale, this comparator subtracts shortfall divided by b_G directly from A. That distinct liquidation rule is another reason its trajectories do not transfer to the two-land model.

B and E use the GFN/NFA convention, with the annual basis implicit. Population P, carrying capacity K, and $K_{\min}$ are in cap; e is in gha cap^{-1}. Debt D is in gha yr, so $\dot{D}$ and ηD are in gha, and α is in gha^{-1} yr^{-1}. Rates ρ, r, η, κ are in yr^{-1}; delays τ_g, τ_p and wave midpoint t_w are in years.

The numerical implementation enforces $A \geq A_{\text{ext}}$, $P, D \geq 0$, and $K \geq K_{\min}$, with the exact positive-part operator. Delayed histories are constant at their initial values, and RK4 integration uses linear interpolation of delayed states. Step sizes are 0.1 yr for the full trajectory (checked at 0.05 and 0.025 yr), 0.01 yr for the fixed-demand mask (with a 0.02 yr shortfall sweep), 0.2 yr for the delay pair (checked at 0.1 yr), and 0.5 yr for the finite-grid classification.

With fixed b, $T_b = 0$, $D = 0$, and $\sigma = 1$, interior equilibria satisfying $0 < A < A_{\max}$ and $B(A)/e > K_{\min}$ obey $E = B(A)$ and $P = B(A)/e$. They form a one-parameter family along

$$
B(A) = (b + b_G\rho)A - \frac{b_G\rho}{A_{\max}}A^2, \qquad B'(A) = b + b_G\rho - 2b_G\rho\frac{A}{A_{\max}}.
$$

An interior maximum within $0 < A < A_{\max}$ requires $b_G\rho > b$. For the debt-free, constant-parameter delay and basin illustrations, $b = 0.5$, $b_G = 0.8$, and $\rho = 0.05$, so $b_G\rho = 0.04 < b$ and $B(A)$ increases throughout that interval. In the full-trajectory illustration, $b(t)$ changes with the technology wave and debt; we make no monotonicity or equilibrium-fold claim for that time-varying case.

For the debt-free reference equilibrium, $A^* = 0.8$ under $(b, b_G, \rho, A_{\max}, e, r) = (0.5, 0.8, 0.05, 1.2, 0.55, 0.02)$. It has $B^* = 0.4107$ gha and $P^* = B^*/e = 0.7467$ cap. The characteristic equation has a neutral root tangent to the equilibrium family and a positive real root. That positive root is $0.5883\,\text{yr}^{-1}$ when $(\tau_g, \tau_p) = (0, 0)$ and $0.6250\,\text{yr}^{-1}$ when $(30, 25)$ yr. These local results apply only to the comparator; they are neither a Hopf classification nor a stability result for the two-land system.

Figures S1–S4 show conditional runs of the comparator. In Fig. S3, the same starting state approaches $A_{\max}$ at $\tau_g = 10$ yr but reaches the imposed floor at $\tau_g = 30$ yr, with $\tau_p = 25$ yr in both cases. This pair shows delay sensitivity for one initial condition; it does not establish a

universal critical delay. Figure S4 classifies a finite grid of starting states at fixed τ_p. Grid-cell proportions are not probabilities over initial conditions or measures of a continuous basin.

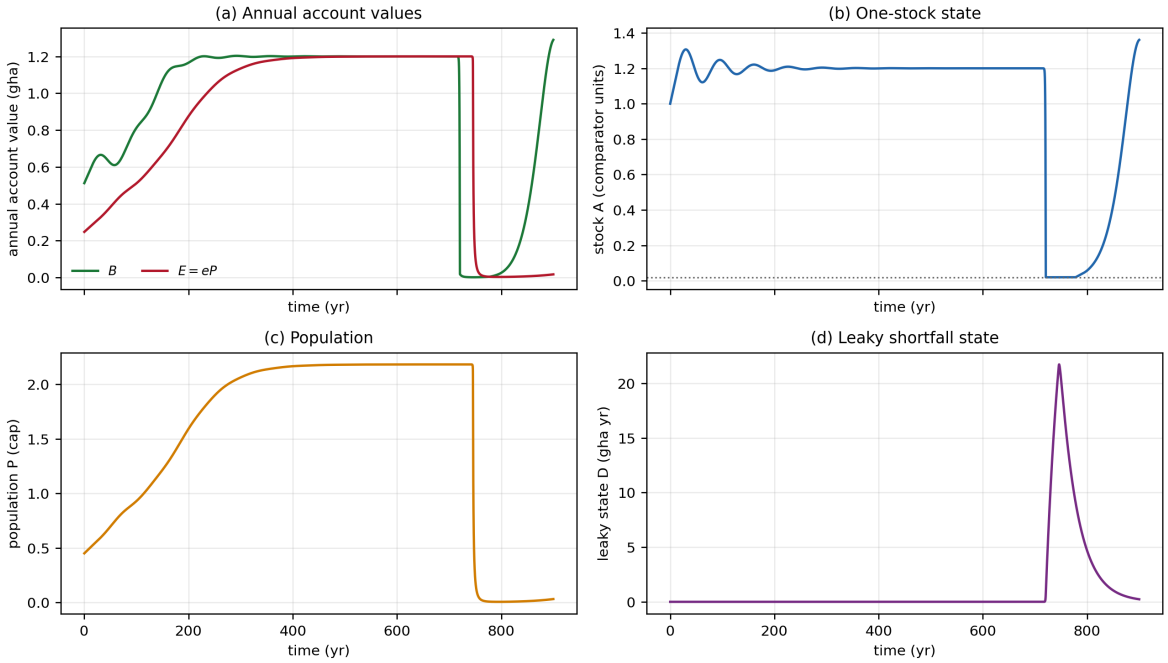


Figure S1: Full delayed one-stock comparator with a logistic technology increment. Parameters are $A_0 = 1.0$ stock unit, $P_0 = 0.45\,\mathrm{cap}$, $b_0 = 0.5$, $b_G = 0.6$, $\rho = 0.08\,\mathrm{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $e = 0.55\,\mathrm{gha\,cap}^{-1}$, $r = 0.02\,\mathrm{yr}^{-1}$, $\eta = 0.03\,\mathrm{yr}^{-1}$, $\alpha = 0.2\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$, $\tau_g = 15\,\mathrm{yr}$, $\tau_p = 25\,\mathrm{yr}$, $A_{\mathrm{ext}} = 0.02$ stock units, $K_{\min} = 10^{-4}\,\mathrm{cap}$, $\Delta b = 0.5\,\mathrm{gha}$ per stock unit, $\kappa = 0.04\,\mathrm{yr}^{-1}$, and $t_w = 120\,\mathrm{yr}$; $\sigma = 1$, $\omega_{Tb} = 1$, and $D_0 = 0$. The positive-part operator is exact. B and E are in gha, A is in comparator-specific stock units, P is in cap, and D is in gha yr. The trajectory reaches the imposed lower bound near 721 yr, then recovers as modeled demand falls; the bound is not absorbing. This is a conditional model illustration, not an observation or a two-land result.

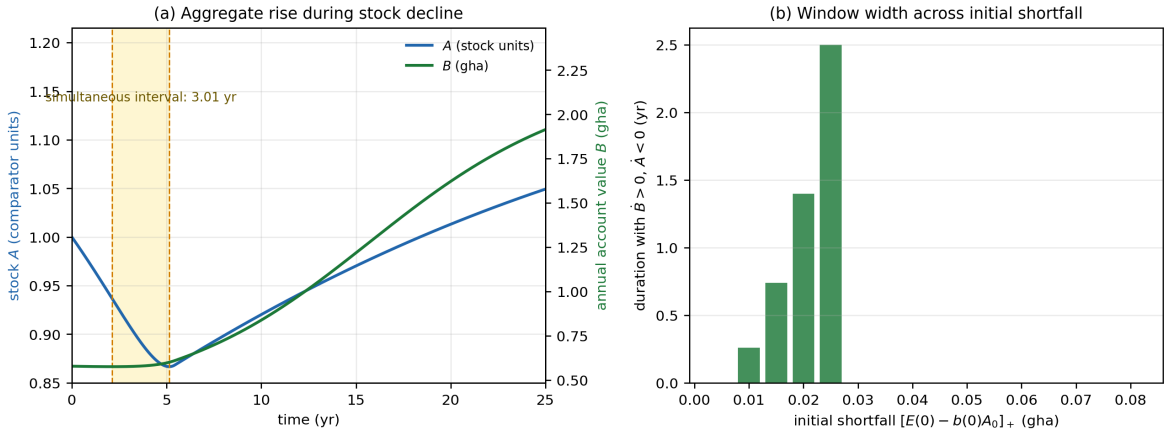


Figure S2: One-stock masking example with fixed demand and no delays. Parameters are $A_0 = 1.0$ stock unit, $E = 0.60\,\mathrm{gha}$, $b_0 = 0.5$, $b_G = 0.8$, $\rho = 0.05\,\mathrm{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $\eta = 0.05\,\mathrm{yr}^{-1}$, $\alpha = 0.03\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$, $\Delta b = 1.5\,\mathrm{gha}$ per stock unit, $\kappa = 0.2\,\mathrm{yr}^{-1}$, and $t_w = 15\,\mathrm{yr}$; $\sigma = 1$, $\omega_{Tb} = 1$, and $D_0 = 0$. With the exact positive-part operator, the interval with $\dot{B} > 0$ and $\dot{A} < 0$ lasts about $3.0\,\mathrm{yr}$. Over that interval, A falls from about 0.937 to 0.867 stock units while B rises from about 0.575 to 0.600 gha. Panel (b) shows the interval width across initial shortfalls $[E(0) - b(0)A_0]_+$ for this parameterization. These are conditional comparator outputs.

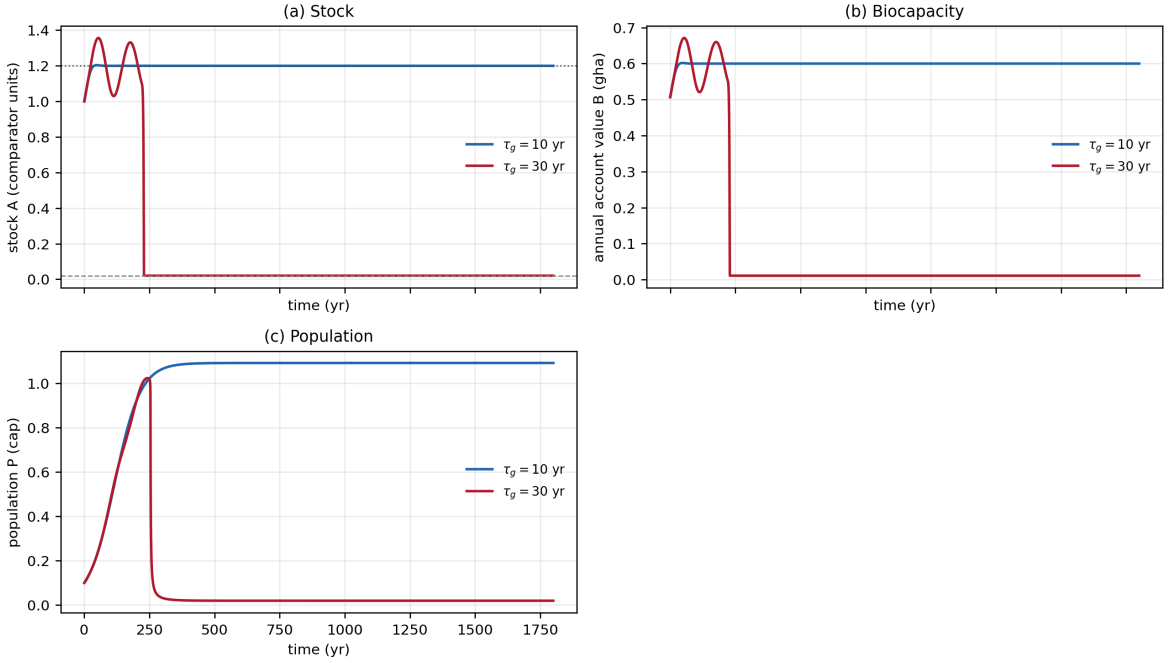


Figure S3: Controlled regeneration-delay examples in the debt-free one-stock subsystem. Both runs use $A_0 = 1.0$ stock unit, $P_0 = 0.1\,\mathrm{cap}$, $b = 0.5$, $b_G = 0.8$, $\rho = 0.05\,\mathrm{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $e = 0.55\,\mathrm{gha\,cap}^{-1}$, $r = 0.02\,\mathrm{yr}^{-1}$, $\tau_p = 25\,\mathrm{yr}$, $A_{\mathrm{ext}} = 0.02$ stock units, $K_{\min} = 10^{-6}\,\mathrm{cap}$, and $\sigma = 1$; only τ_g differs (10 versus 30 yr). B is in gha, P in cap, and A in comparator-specific stock units. These selected trajectories do not establish a general critical delay.

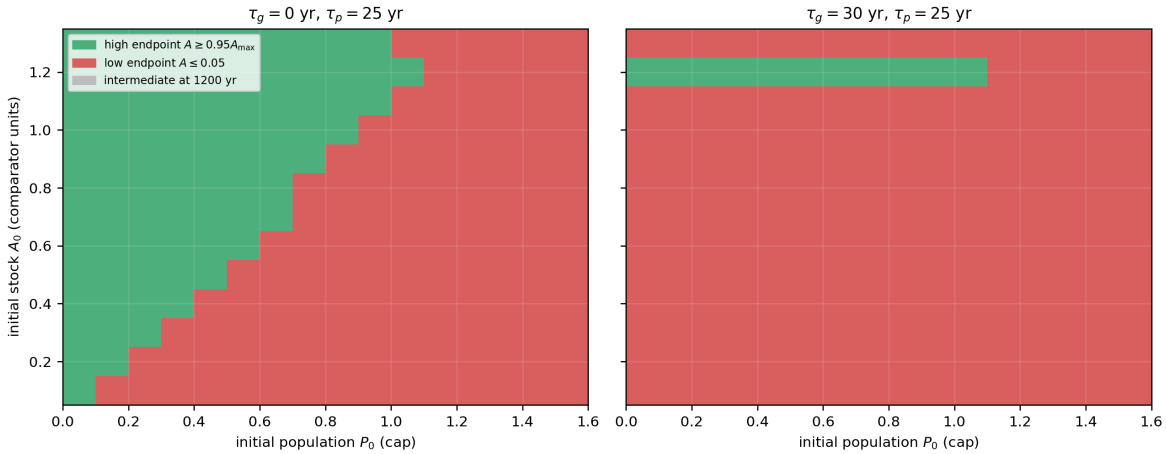


Figure S4: Finite-grid initial-condition classification in the debt-free one-stock subsystem. The runs use the parameter set in Fig. S3, with $K_{\min} = 10^{-6}$ cap and $\tau_p = 25$ yr; $\tau_g = 0$ yr in the left panel and 30 yr in the right. The imposed lower bound is $A_{\text{ext}} = 0.02$ stock units. The grid uses $A_0 = 0.10, 0.20, \ldots, 1.30$ stock units and $P_0 = 0.05, 0.15, \ldots, 1.55$ cap; trajectories run to 1200 yr with $\Delta t = 0.5$ yr. Green marks a high endpoint $A(1200) \geq 0.95A_{\max}$, red a low endpoint $A(1200) \leq 0.05$ stock units, and gray an intermediate endpoint. These display-grid categories are not probabilities or continuous-basin measures.

S8 Controlled technology-wave experiment

The matched experiment compares endogenous-demand runs with the technology increment T_b switched on or off, at $P_0 = 1.0$ or 1.2 cap. Within each pair, the runs share $A_{tot} = 2.5$ ha, $A_{f0} = 0.6$ ha, $A_{c0} = 0.9$ ha, $q_0 = 0.9$, and $D_0 = 0.0$ gha yr. Parameters and numerical methods are matched; only ω_{Tb} differs. Both runs also share constant pre-run histories at $q_0 = 0.9$ and the no-wave initial account value $B = 0.560184$ gha.

Shared parameter values are $b_{f0} = 0.85$ gha ha^{-1}, $b_c = 0.05$ gha ha^{-1}, $b_{G,c} = 0.8$ gha yr ha^{-1}, $\rho_c = 0.08$ yr^{-1}, $\alpha = 0.03$ gha^{-1} yr^{-1}, $\eta = 0.05$ yr^{-1}, $\eta_f = 0.05$ yr^{-1}, $\mu = 0.06$ yr^{-1}, $e = 0.55$ gha cap^{-1}, $r = 0.02$ yr^{-1}, $\tau_g = 20$ yr, $\tau_p = 25$ yr, $\tau_{conv} = 1$ yr, and $\chi_r = 0.10$ gha^{-1} yr^{-1}. The active wave uses $\Delta b = 1.0$ gha ha^{-1}, $\kappa_w = 0.1$ yr^{-1}, and $t_{wave} = 40$ yr; runs last 250 yr. The logistic increment is already nonzero at $t = 0$ ($T_b(0) = 0.017986$ gha ha^{-1}), so initial B is 0.560184 gha without the wave and 0.570976 gha with it. This small difference comes from the logistic tail. Land, quality, population, and debt states otherwise match, and the pre-run B history remains common. The wave reaches half its increment at $t_{wave} = 40$ yr.

Figure S5 shows the $P_0 = 1.0$ pair. In both runs, peak debt occurs after the wave midpoint: $D_{max} = 0.008126$ gha yr at 86.15 yr without the wave and 0.020205 gha yr at 115.60 yr with it. At $P_0 = 1.2$, the all-time maximum occurs early and is lower with the wave, but the maximum for $t \geq t_{wave}$ is higher. Table S13 reports both windows. Because t_{wave} marks the logistic midpoint, the post-midpoint window includes the wave's continued rise.

For $P_0 = 1.0$, the all-time wave/no-wave peak-debt ratio is 2.487 at $\Delta t = 0.05$ yr and 2.492 at 0.025 yr; under current-quality regeneration it is 2.422 at 0.05 yr. For $P_0 = 1.2$, all-time ratios are 0.834 and 0.833, while post-midpoint ratios are 2.320 and 2.328. Halving the step changes debt maxima by less than 0.9%. In the $P_0 = 1.0$ pair, A_c stays at or above its initial 0.9 ha, so this experiment illustrates a conditional delayed debt rebound, not the frozen-demand capital-land drawdown. At $P_0 = 1.2$, the all-time and post-midpoint maxima tell different stories. The debt variable D is a leaky state driven by aggregate deficit $E - B$, not an observed NFA or physical-stock quantity.

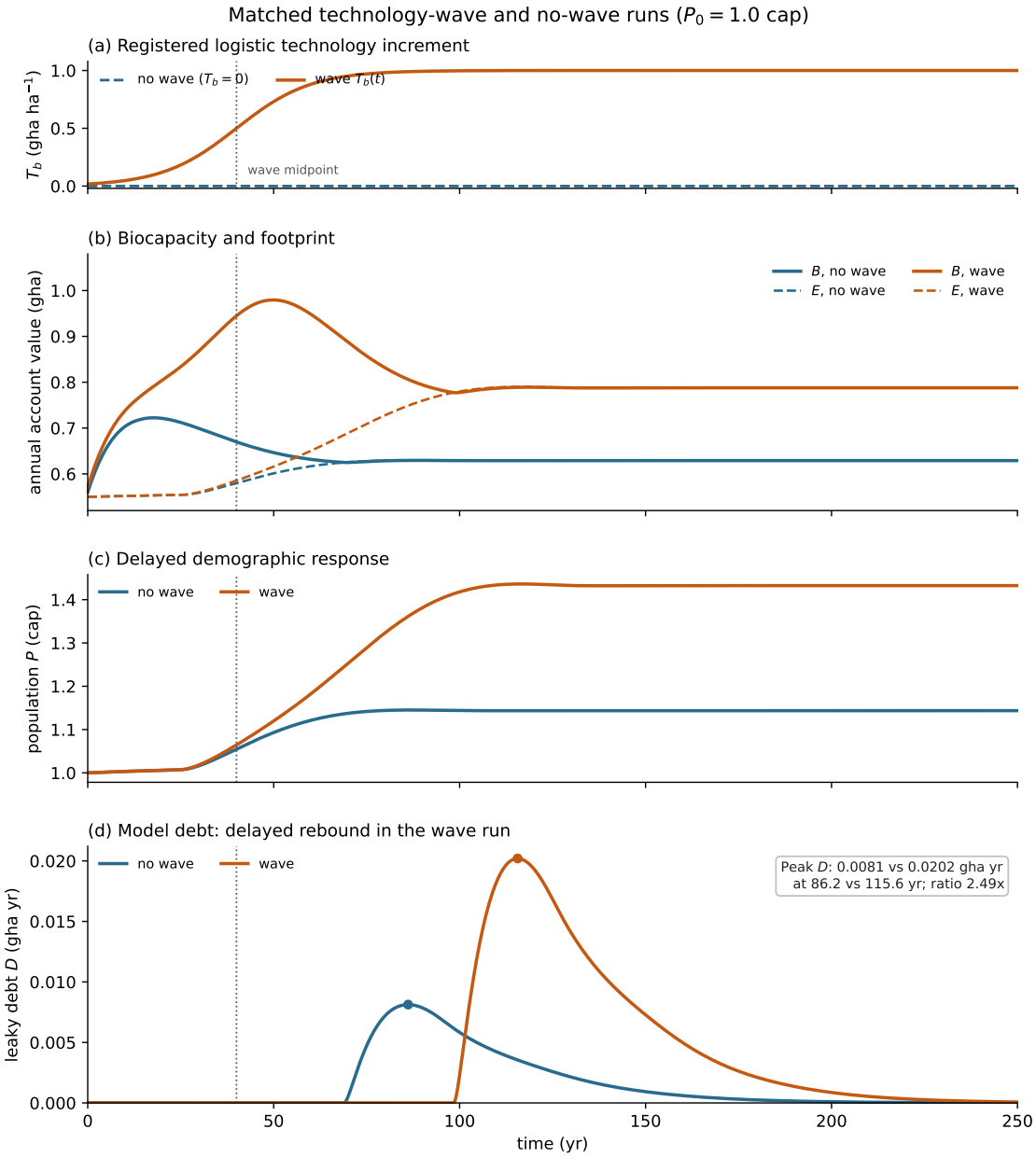


Figure S5: Matched endogenous-demand technology-wave/no-wave simulation at $P_0 = 1.0$ cap. Parameters and initial conditions are in SI §S8; only T_b is switched on or off. (a) The active logistic increment $T_b(t)$, with the dotted line at its midpoint $t_{wave} = 40$ yr; (b) annual account values B and E (gha); (c) population P (cap); and (d) leaky model debt D (gha yr). Markers denote all-time maxima: 86.15 yr without the wave and 115.60 yr with it; peak debt is 0.008126 versus 0.020205 gha yr. The wave begins with its small logistic tail at $t = 0$ and approaches $\Delta b = 1.0$ gha ha^{-1}. The dotted line marks the midpoint, not an end date.

Table S13: Matched technology-wave/no-wave debt maxima. Entries are D_{max} in gha yr, with the time of the maximum in years in parentheses; the ratio is wave/no-wave. "All" is the maximum over $0 \leq t \leq 250$ yr; $t \geq 40$ is the post-midpoint maximum. The $P_0 = 1.0$ all-time peaks both occur after the midpoint.

P_0 (cap)	quality update	Δt (yr)	window	no wave D_{max} (time)	wave D_{max} (time)	ratio
1.0	delayed	0.050	all	0.008126 (86.15)	0.020205 (115.60)	2.487
1.0	delayed	0.025	all	0.008105 (86.20)	0.020195 (115.65)	2.492
1.0	current	0.050	all	0.008326 (85.70)	0.020168 (115.60)	2.422
1.2	delayed	0.050	all	0.061218 (1.65)	0.051037 (1.45)	0.834
1.2	delayed	0.050	$t \geq 40$	0.010854 (81.70)	0.025184 (112.55)	2.320
1.2	delayed	0.025	all	0.060732 (1.625)	0.050582 (1.475)	0.833
1.2	delayed	0.025	$t \geq 40$	0.010808 (81.775)	0.025165 (112.625)	2.328

S9 Additional demographic-delay scan

The main text varies regeneration delay τ_g; this additional scan varies the demographic delay τ_p in the canonical delayed-quality solver. It uses the baseline parameter set, $\tau_g = 20$ yr, no restoration ($\chi_r = 0$), and the typed floor $A_c^{\min} = 0.05$ ha.

For a finite-grid check, we cross four initial capital areas ($A_{c0} = 0.40, 0.70, 1.00, 1.18$ ha) with six initial populations ($P_0 = 0.5, 1.0, 1.5, 2.0, 2.5, 3.0$ cap). At $\tau_p = 25, 100, 200, 300$, and 500 yr, 20 of the 24 runs do not reach the floor-proximity criterion $\min_t A_c \leq 0.05 + 10^{-6}$ ha within 1500 yr. This is a count on the stated grid and horizon, not a probability or a general basin measure.

A second check follows one endogenous-demand trajectory for 3000 yr at $\Delta t = 0.1$ yr, with $A_{c0} = 0.9$ ha, $A_{f0} = 0.6$ ha, $q_0 = 0.9$, $P_0 = 1.2$ cap, and $D_0 = 0$. We divide the run into four equal 750-year windows and report the population range within each. As τ_p lengthens, the largest range occurs in a later window; subsequent windows are nearly flat in this trajectory. No repeated oscillation appears within the simulated horizon. This finite trajectory check does not prove the absence of a Hopf bifurcation or of cycles under other conditions.

Table S14: Population range $\max(P) - \min(P)$ within each 750-year window of the selected delayed-quality runs. Population is in cap; all runs use $T = 3000$ yr and $\Delta t = 0.1$ yr. The four window ranges are descriptive summaries, not stability criteria.

τ_p (yr)	0–750	750–1500	1500–2250	2250–3000
25	0.3681	0.0000	0.0000	0.0000
300	0.4699	0.0001	0.0000	0.0000
500	0.4652	0.0052	0.0000	0.0000
1000	0.1815	0.4703	0.0000	0.0000
1500	0.1815	0.0000	0.4704	0.0000
2000	0.1815	0.0000	0.4652	0.0052

S9.1 Selected two-land reference values

Table S15 gathers values used across the main text and supplement. The model values are conditional on the representative baseline parameterization; the floor-proximity row is numerical and tolerance-defined.

Table S15: Selected two-land reference values for cross-checking. Values are conditional model results, not observations of physical stocks. The initial-population values are proximity markers for the stated floor tolerance.

Quantity	Value	Location
Composition diagnostic at $q^* = q_{\max}/2$	$\Delta b_{\text{conv}}(q^*) = +0.784\,\text{gha ha}^{-1}$	S4.1; fast-land yield $b_f = 0.85\,\text{gha ha}^{-1}$
Reduced fixed-demand ceiling	$E_{\text{ceil}} = 1.138\,\text{gha}$	S2.2
Deficit-side eigenvalues	$\lambda_1 = -0.0537$, $\lambda_2 = -0.998\,\text{yr}^{-1}$	S2.2
Crossover shortfall	$S^{\dagger} = 1.296\,\text{gha}$	S2.5
Debt fixed points	$D^* = 1.186\,\text{gha yr}$ at $E = 1.158\,\text{gha}$; $D^* = 5.375\,\text{gha yr}$ at $E = 1.238\,\text{gha}$	S2.4
P_0 within 10^{-6} ha of the floor	2.41, 2.70, 2.98, 3.14 cap for $A_{c0} = 0.40,\ 0.70,\ 1.00,\ 1.18\,\text{ha}$	S4.2; $\Delta t = 0.05\,\text{yr}$, $T = 1000\,\text{yr}$

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